

ReactWithSummary[@T][k]

ROLE: SYSTEM

♦ TASK_DESC // you are an assistant who's job is...

♦ AVAILABLE_TOOLS

== History ==

◇ If ($@T - k > 0$):

ROLE: ASSISTANT

♦ THIS_IS_A_SUMMARY

Summarize($\text{prompt.History}[@T - k - 1]$)

⌋ ForEach $t : \max(1, @T - k) \dots @T - 1$

ROLE: USER

$\text{env.user_question}[@t]$

⌋ ForEach $i : 1 \dots \text{sys.substeps}[@t]$

ROLE: ASSISTANT

$\text{resp.tool_reasoning}[@t.i]$

$\text{sys.tool_used}[@t.i]$

ROLE: TOOL

$\text{sys.tool_used}[@t.i].\text{tool_response}$

ROLE: USER

$\text{env.user_question}[@T]$

⌋ ForEach $i : 1 \dots \text{sys.substeps}[@T]$

ROLE: ASSISTANT

$\text{resp.tool_reasoning}[@T.i]$

$\text{sys.tool_used}[@T.i]$

ROLE: TOOL

$\text{sys.tool_used}[@T.i].\text{tool_response}$

== End History ==

ROLE: SYSTEM

♦ INSTRUCTION // choose an action, use ReACT

